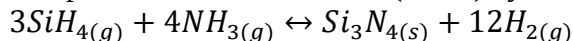


MSE 360: Kinetic Processes in Materials

Homework #9

Instructions: Answer the following questions on a separate piece of paper. Be sure that your name and date are on the first page. Upload your work as a PDF to Moodle for credit. *All work must be done individually.*

1. Consider the chemical vapor deposition of silicon nitride (Si_3N_4) by the following reaction:

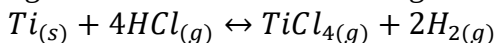


- a) On the same graph, make a schematic plot of the natural logarithm of the Si_3N_4 growth rate versus $1/T$ (K^{-1}) for each of the following three growth processes and identify each on the plot:
- A surface-reaction controlled process
 - A gas diffusion controlled process
 - The actual rate-limiting rate when both are occurring in series
- b) Explain why each plot has the temperature dependence shown.
- c) Make a plot of the SiH_4 gas pressure as a function of distance away from the growing silicon nitride film surface for the following three cases:
- Surface reaction control
 - Gas diffusion control
 - When both processes have about the same rate

Clearly identify each case and indicate important distances and pressures. Assume that the gases are flowing over a flat substrate on which the deposition is being made.

2. A vacuum reactor is used to synthesize WO_3 nanoparticles for use in an electrochromic window device. To create the WO_3 nanoparticles, a tungsten (W) metal filament 1.0 mm in diameter and 10 cm long is heated to $2,200^\circ\text{C}$ in a vacuum with a trace pressure of oxygen gas. Assuming all of the W evaporating from the surface of the filament is rapidly and completely oxidized to WO_3 , determine the maximum rate of production of WO_3 from this reactor (g/s). The equilibrium vapor pressure of W at $2,200^\circ\text{C}$ is approximately 10^{-10} atm.

3. Titanium undergoes active gas corrosion in HCl according to the reaction



The rate of etching of the Ti surface is found to be $2.88 \mu\text{m}/\text{min}$. The following additional information is provided:

- $k_0 = 1.2 \times 10^2 \text{ cm/s}$
- $P_{\text{HCl}} = 0.10 \text{ atm}$
- $M_{\text{Ti}} = 48 \text{ g/mol}$
- $\rho_{\text{Ti}} = 4.5 \text{ g/cm}^3$
- $T = 1327^\circ\text{C}$

- Calculate the rate constant for the reaction.
- Calculate the activation energy for the reaction.
- What is the time required to remove 5 mm of Ti?

d) What is the corrosion rate after a time $t = 20$ h?